

TER Proposal: Comparing Finetuning and Few-shot Approaches for Large Language Models applied to Natural Language Inference for Clinical Trials

November 18, 2025

1 Context

The intern will be working in the SEME group on the CTInfer project. The CTInfer project aims to study and to develop techniques for applying *natural language inference* (NLI) to *clinical trials* (Aguiar et al., 2025, 2024). Natural language inference consists of deciding whether a statement entails or contradicts a given premise. In a clinical context, this can be useful for recruiting patients in clinical trials by matching their information with the inclusion and exclusion criteria of a clinical trial (Aguiar et al., 2025) (see Fig. 1).

2 Objectives

The primary objective of this TER project is to compare fine-tuning and few-shot approaches for large language models (LLMs) applied to natural language inference for clinical trials.

This TER project will use two datasets to apply NLI on clinical trials: NLI4CT (Jullien et al., 2023) and NLI4PR (Aguiar et al., 2025). NLI4CT applies NLI to breast cancer trials by covering various sections of the trial report: eligibility, intervention, results, and adverse events. The dataset encompasses various types of inference and knowledge, including numerical, logical, biomedical, and general knowledge. NLI4PR is centered on determining the potential eligibility of a considered patient for the trial. However, unlike NLI4CT, NLI4PR uses patient profiles described using patient language instead of the doctor’s medical language.

In previous work, we conducted experiments in a zero-shot setting, using models without further fine-tuning. In this work, we aim to investigate whether using different finetuning techniques can improve the models’ performance. Indeed, LLMs have been demonstrating a boost of performance when finetuned on some specialized tasks (Wu et al., 2023; Labrak et al., 2024; Chen et al., 2023). To do so, we propose to use the training sets of NLI4CT and NLI4PR to finetune LLMs, such as Qwen (Yang et al., 2024), which performs well on these tasks. The first approach can be based on a simple supervised finetuning and, more complex approaches, such as model merging, can be further explored. The experiments could also use LoRA (Hu et al., 2021) to efficiently finetune the considered LLMs.

The results will be compared with a set of baselines (approaches using few-shot examples in the prompt).

3 Tasks

The main tasks are the following:

1. Literature review of the different finetuning techniques.
2. Finetuning of LLMs.

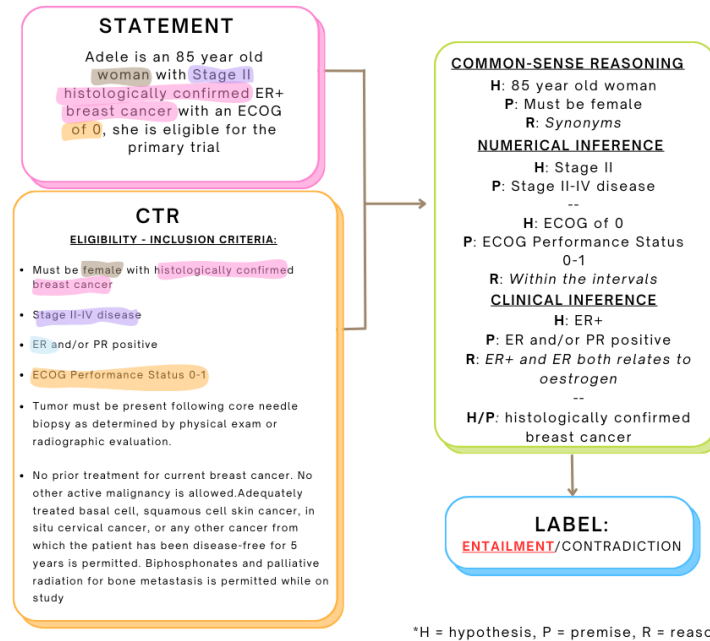


Figure 1: Examples of Natural Language Inference on a Clinical Trial

3. Implementation of optimized few-shot prompts.
4. Quantitative and qualitative evaluation of the obtained results and comparison with a set of baselines.

4 Technical stack

The chosen programming language is Python; however, the intern will decide on the frameworks or libraries based on their specific needs.

5 Your skills

Good programming skills in Python and PyTorch are required. Experience using libraries such as Pandas, SpaCy, and NLTK is preferable. You will work collaboratively with other researchers and must have good writing and spoken English skills.

6 General information

The TER project will be carried out at Laboratoire Interdisciplinaire des Sciences du Numérique (LISN) located at Campus Universitaire bâtiment 507, Rue du Belvédère, 91400 Orsay.

Supervisors: Nona Naderi (naderi@lisn.fr), Pierre Zweigenbaum (pz@lisn.fr), Mathilde Aguiar (mathilde.aguiar@lisn.fr)

Duration: 2-4 months (with possible extension for M1 internship)

Start date: Jan 2026

Candidates can apply by sending an email to the TER supervisors.

References

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